

State Labor-Market Data

Core Data-Integrity Validation

Structure • Calendar • Release timing • Revisions (modules V1-V4)
Continued claims • U3 unemployment • Nonfarm payrolls — 50 states + DC, FRED/ALFRED

ALL THREE DATASETS CLEARED on the core modules (V1-V4)

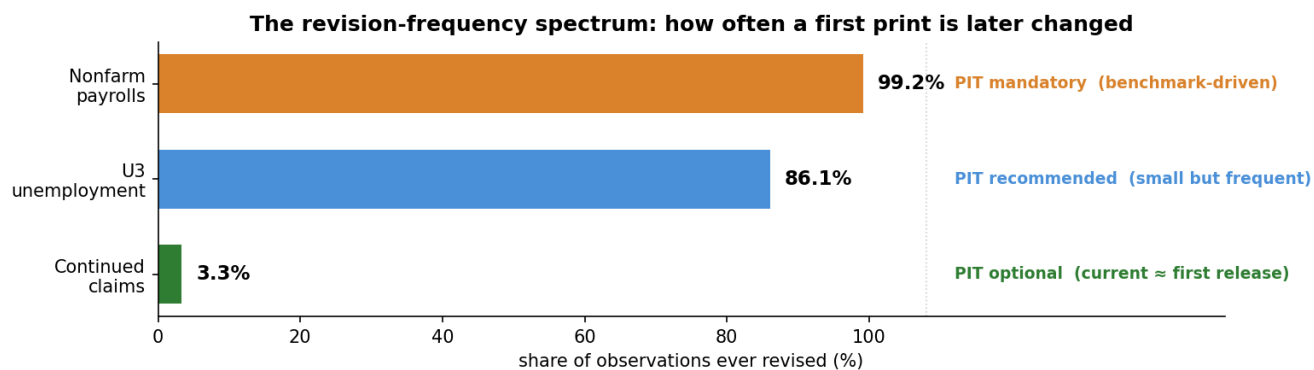
	Checks	PASS	WARN	FAIL
All datasets	55	38	17	0
Continued claims	21	13	8	0
U3 unemployment	17	12	5	0
Nonfarm payrolls	17	13	4	0

Scope. This concise report covers only the core data-integrity modules - structural integrity (V1), calendar & frequency (V2), release calendar & point-in-time boundary (V3), and revision behaviour (V4) - applied identically to all three series. The three mechanical point-in-time guards (leak check, availability-frontier monotonicity, look-ahead guard) and the downstream modules (outliers, cross-section, seasonality, lead-lag, backtest readiness) are covered in the full per-dataset suites.

Cross-dataset summary

On the core modules the three series are structurally clean and share one publication calendar (U3 and NFP in the same monthly report; claims weekly). They differ almost entirely on **V4 - revisions**, which is the operative distinction for point-in-time modelling:

	Continued claims	U3 unemployment	Nonfarm payrolls
V1 data type	weekly count	bounded rate (pp)	monthly count
V2 calendar	weekly, Sat week-end	monthly (Oct-25 hole)	monthly
V3 vintage start	Sep 2010	2005-07 (ramp to 2008)	2005-07 (ramp to 2008)
V3 publication lag	~13-19 days	~50 days	~50 days
V4 share revised	~3% (rare)	~86% (0.1pp-quantized)	~99% (benchmark-driven)
V4 PIT requirement	optional	recommended	mandatory



V4 across the three series: share of observations ever changed after first release. This single axis sets how strictly point-in-time discipline must be enforced - optional for claims, mandatory for NFP.

Continued claims

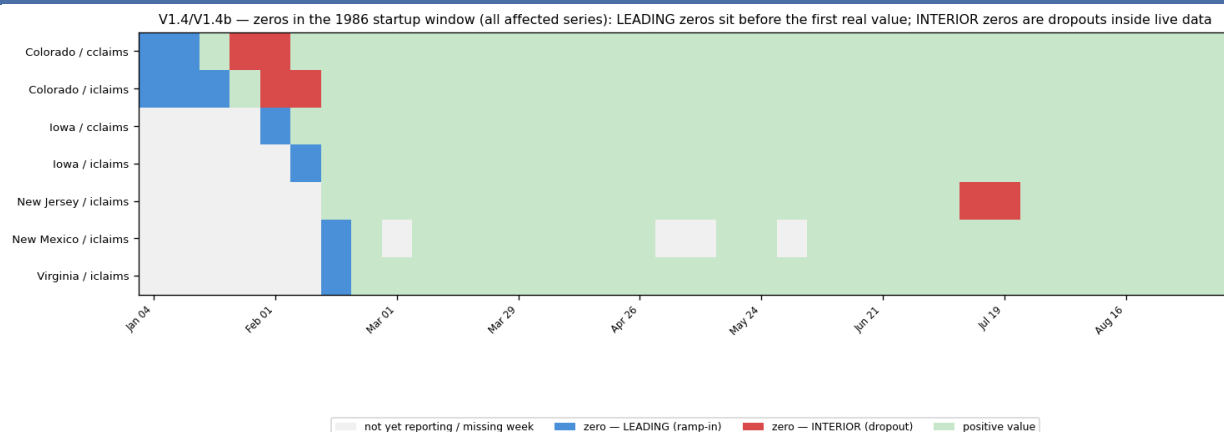
{ABBR}CCLAIMS - weekly, NSA — 13 PASS / 8 WARN / 0 FAIL on 21 core checks

The timely, barely-revised weekly series. Structure and calendar are clean (weekly Saturday week-ends); the release archive begins Sep 2010 with a July-2020 capture-regime switch; only ~3% of weeks are ever revised.

ID	Check / question	Res	Metric
V1 - Structural integrity			
V1.1	Do we have all 51 states in every table?	PASS	{'vintage': 51, 'current': 51, 'pit': 51}
V1.2	Is (state, claim_type, week) a unique key?	PASS	vintage=0, current=0, pit=0
V1.3	Are key fields complete (no nulls)?	PASS	0 nulls
V1.4	Is every value parseable, finite and positive?	WARN	vintage bad=0; current zeros=15 (latest 1986-07-19), post-1990=0
V1.4b	Are zeros ramp-in artifacts (leading) or mid-sample dropouts (interior)?	WARN	9 leading, 6 interior; interior post-1990: 0
V1.5	Did any state's fetch silently truncate?	PASS	min=2100, max=2112
V2 - Calendar & frequency			
V2.1	Do obs dates follow the DOL Saturday week-ending convention?	PASS	0 non-Saturday
V2.2a	Are there missing weeks in the pre-1990 history?	WARN	48 missing weeks across 22 states (1986-88)
V2.2b	Are there missing weeks in modern data (live issues)?	PASS	none
V2.3	Do all states cover the same window?	PASS	start spread=5w, end spread=1w
V3 - Release calendar & point-in-time boundary			
V3.1	How far back can point-in-time be reconstructed?	WARN	first vintage 2010-09-16
V3.2	Does the vintage calendar match the DOL mid-week release schedule?	PASS	Thu 96.7%, Wed 3.3% of 867 vintages
V3.3	Is the national first release the advance print at obs+12d?	PASS	median lag 12d (min 11, max 61)
V3.4	On which weekday does FRED capture state data?	PASS	Mon 4.9%, Tue 0.1%, Wed 1.3%, Thu 60.1%, Fri 33.5%
V3.5	How long after the public release do state values appear?	PASS	32.5% within 0-2d of national release; 61.1% at +6-8d (next report)
V3.5b	Is the state-vs-national gap pattern stable within each era?	WARN	pre-2020: 97.8% at +6-8d; post-2020: 83.3% at +0-2d (+13.0% slip to +3-5d = Monday ingestion)
V3.7	Has FRED's capture timing / stored print type been stable?	WARN	switch ~2020-07-10: median 19d -> 13d
V4 - Revision behaviour			
V4.1	How often does a first print get revised at all?	WARN	1403 of 42,124 obs revised (3.33%)
V4.1b	Where are revisions concentrated?	WARN	top-4 states = 56% of exceptions: Hawaii n=404 (med [0.2%]), Pennsylvania n=168 (med [0.1%]), New Mexico n=111 (med [3.0%]), Maryland n=104 (med [0.1%])
V4.2	Do values change mid-life and net out (invisible to endpoint checks)?	PASS	core states: 49 multi-valued obs of 10,556 (0.46%); Hawaii: 405 of 2152 (up to 5 values per obs)
V4.3	How big is the national advance-to-revised gap?	PASS	median -0.09%, IQR [-0.39%, +0.21%], max [6.4%]

Core diagnostics

V1 - Structural integrity



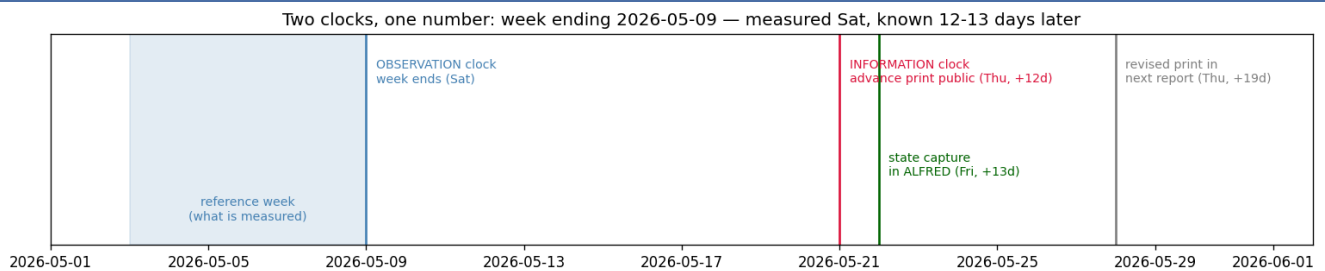
claims_fig_01. V1.4/V1.4b — zeros in the 1986 startup window (all affected series):

V2 - Calendar & frequency

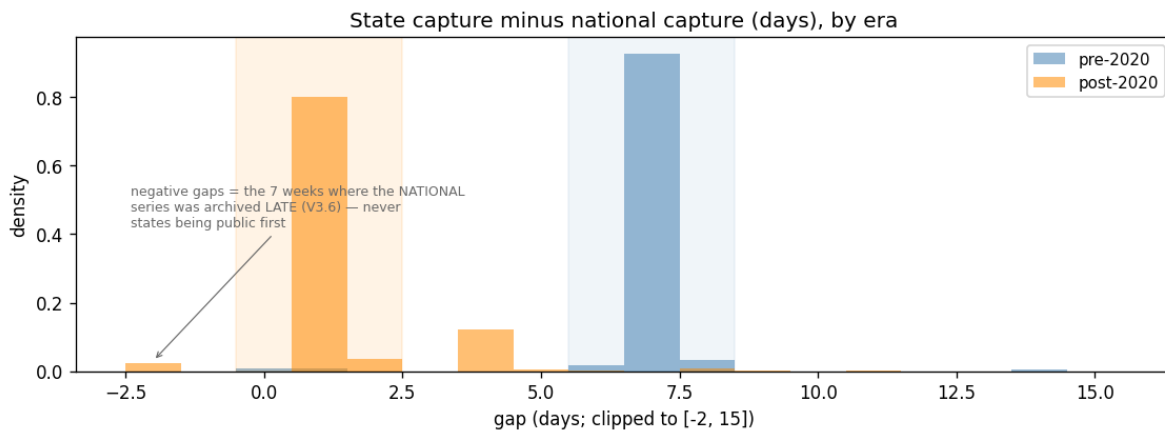


claims_fig_02. Data holes by state — OBSERVATION clock: reference weeks (obs_date) missing from current history

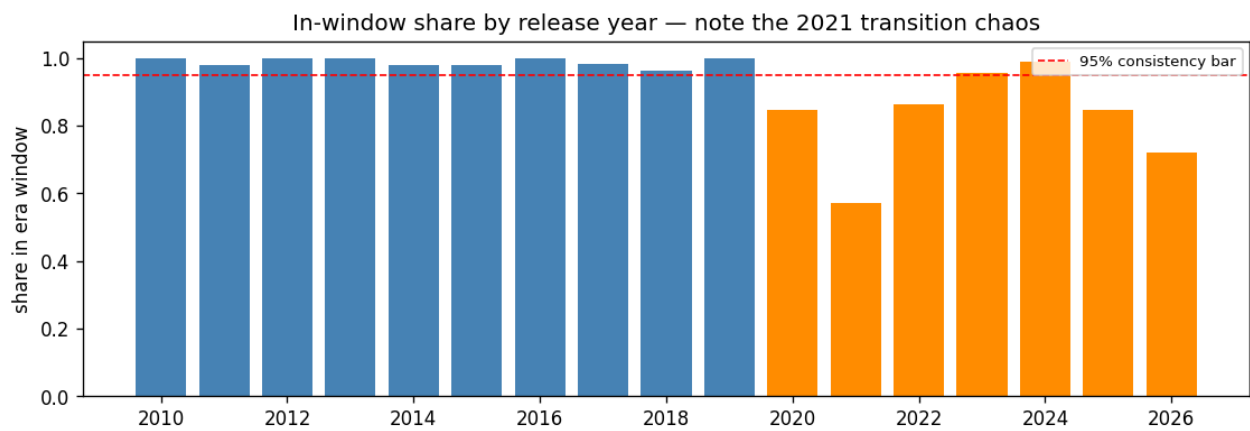
V3 - Release calendar & point-in-time boundary



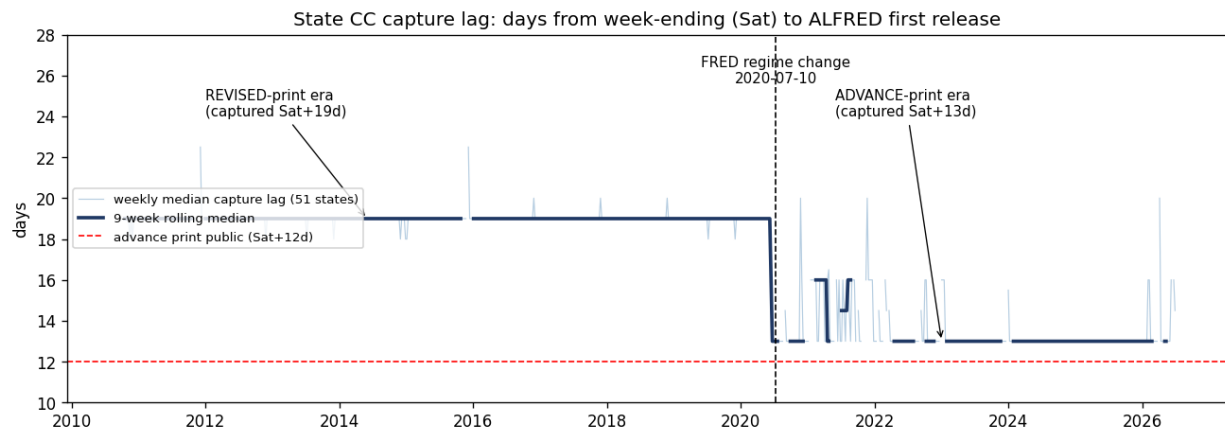
claims_fig_03. Two clocks, one number: week ending the reference week — measured Sat, known 12-13 days later



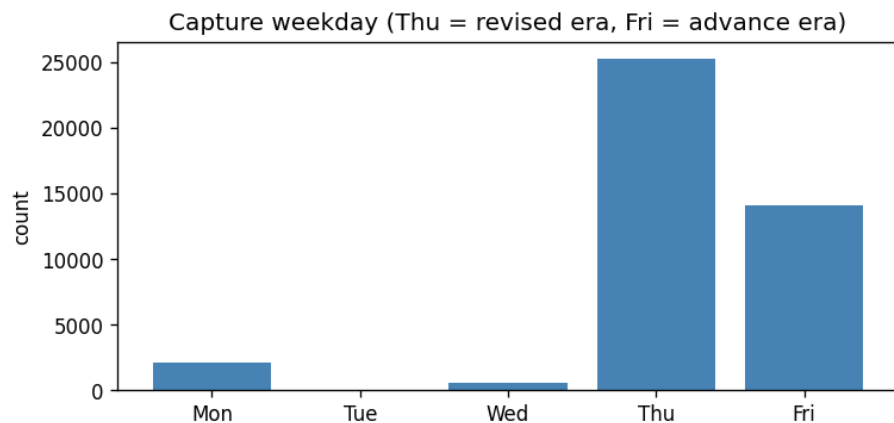
claims_fig_04. State capture minus national capture (days), by era



claims_fig_05. In-window share by release year — note the 2021 transition chaos

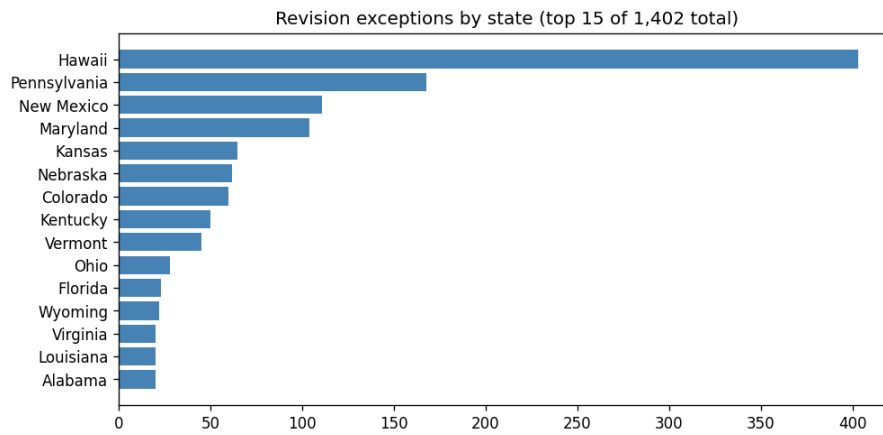


claims_fig_06. State CC capture lag: days from week-ending (Sat) to ALFRED first release

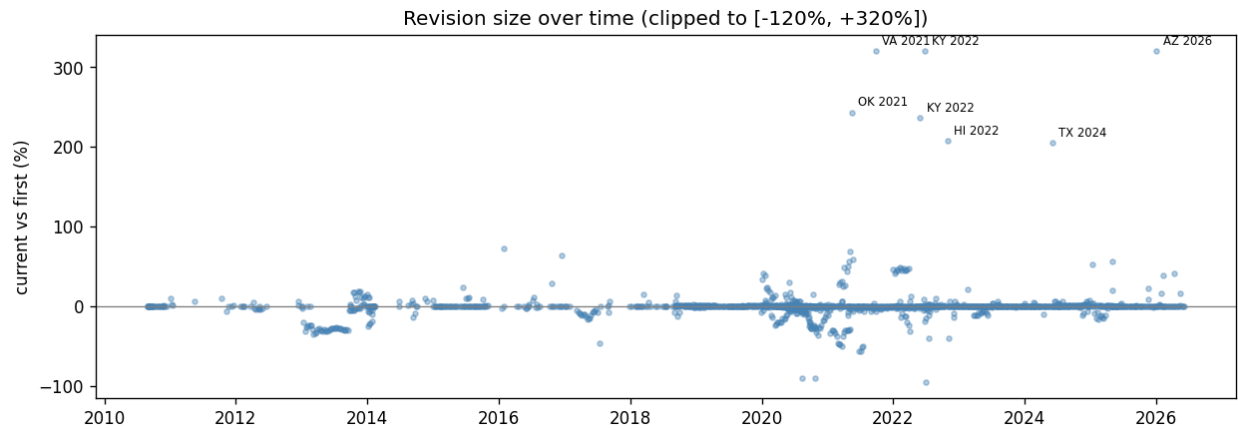


claims_fig_07. Capture weekday (Thu = revised era, Fri = advance era)

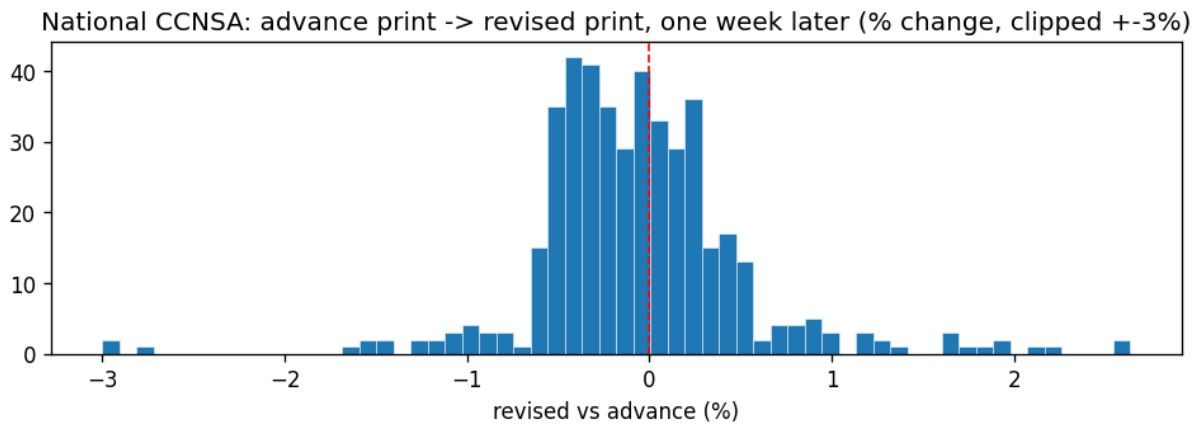
V4 - Revision behaviour



claims_fig_08. Revision exceptions by state (top 15 of 1,402 total)



claims_fig_09. Revision size over time (clipped to [-120%, +320%])



claims_fig_10. National CCNSA: advance print -> revised print, one week later (% change, clipped +-3%)

U3 unemployment

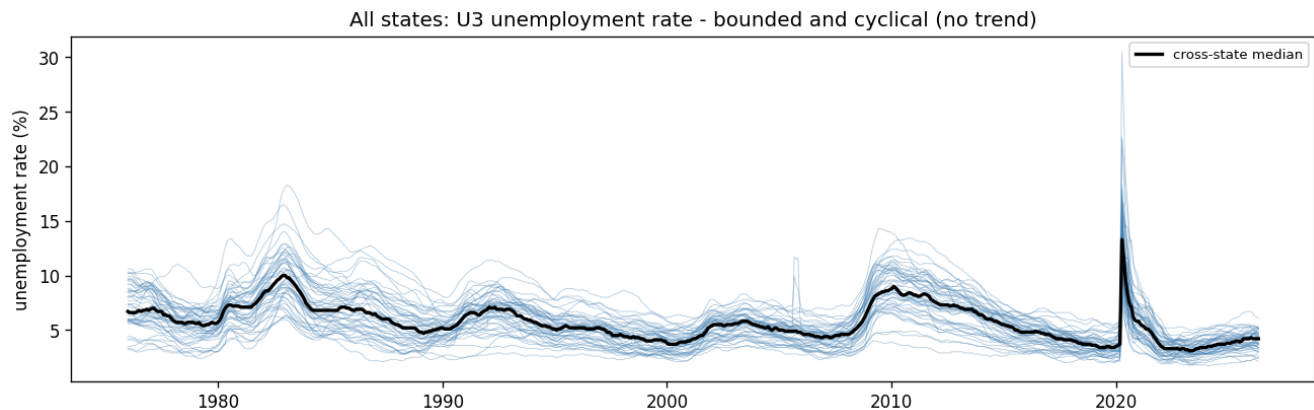
{ABBR}UR - monthly, SA rate — 12 PASS / 5 WARN / 0 FAIL on 17 core checks

A bounded monthly rate (values 1.7-30.5%, quantized to 0.1pp). Calendar is a clean monthly panel with one systematic hole (Oct-2025, the federal shutdown); ~86% of months change but most within the rounding.

ID	Check / question	Res	Metric
V1 - Structural integrity			
V1.1	Do we have all 51 states in every table?	PASS	{'vintage': 51, 'UR current': 51, 'pit': 51}
V1.2	Is (state, series, month) a unique key?	PASS	vintage=0, current=0, pit=0
V1.3	Are key fields complete (no nulls)?	PASS	0 nulls
V1.4	Are all rates in a plausible bounded range?	PASS	0 out-of-range; observed [1.7, 30.5]%
V1.5	Are rates quantized to 0.1pp as published?	PASS	all values 1-decimal: True
V2 - Calendar & frequency			
V2.1	Are all observation dates month-starts?	PASS	0 non-first-of-month
V2.2	Are there missing months in any state?	WARN	1 month(s) missing across all states: 2025-10
V2.3	Do all states cover the same window?	PASS	start 1976-01-01..1976-01-01, end 2026-05-01..2026-05-01
V3 - Release calendar & point-in-time boundary			
V3.1	How far back can point-in-time be reconstructed?	WARN	first vintage 2005-07-22
V3.2	Does the vintage calendar match the jobs-report schedule?	PASS	Mon 2%, Tue 3%, Wed 4%, Thu 7%, Fri 84%
V3.3	Is the state publication lag ~50 days?	PASS	median 50d (IQR 48-53)
V3.4	Do state releases follow the national jobs report?	PASS	100.0% of state months released after national; median gap +14d
V3.5	Is the publication-lag regime stable over time?	WARN	median lag by year spans 48-66d (range 18d)
V4 - Revision behaviour			
V4.1	How often/how much does a first print get revised?	WARN	86% of months revised; exact-zero 14%, within +/-0.1pp 25%, within +/-0.5pp 84%, max 4.1 pp
V4.2	Do larger revisions cluster at the benchmark?	PASS	99% of largest revisions land in Jan-Apr; modal release month = 3 (1581 of 1656 obs across CA/TX/NY)
V4.3	Is there a systematic bias in first prints?	WARN	mean revision -0.047pp (first prints overstate on average)
V4.4	How large is the national revision (calibration)?	PASS	median +0.00pp, IQR [+0.00, +0.10]pp, max 0.4 pp

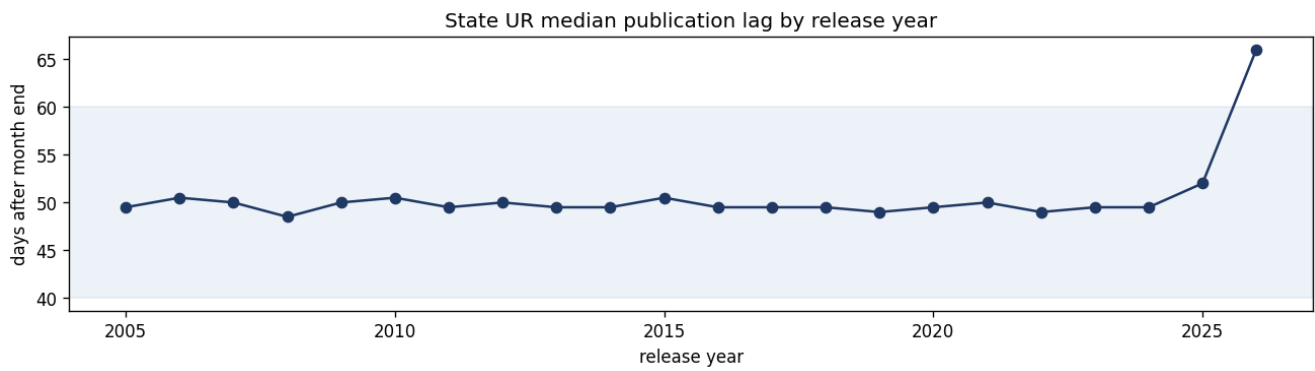
Core diagnostics

V1 - Structural integrity

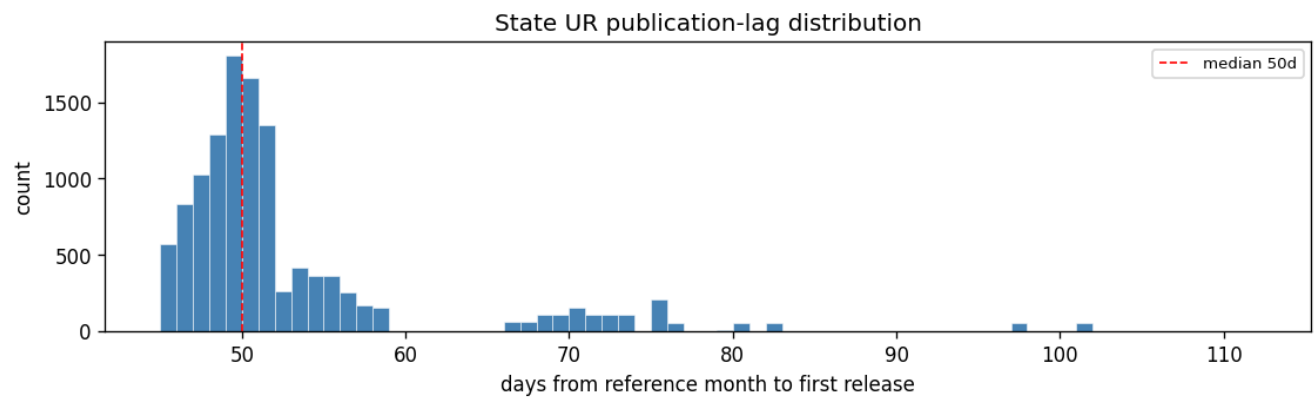


ur_fig_13. All states: U3 unemployment rate - bounded and cyclical (no trend)

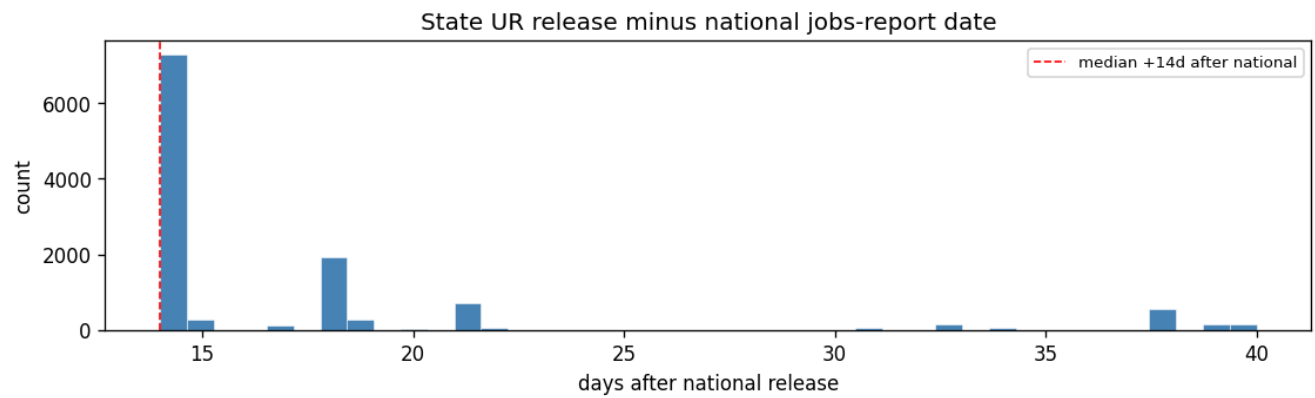
V3 - Release calendar & point-in-time boundary



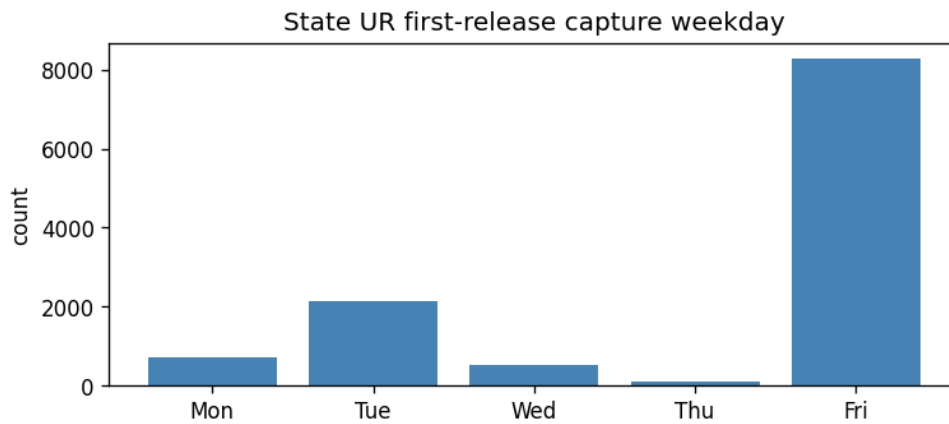
ur_fig_01. State UR median publication lag by release year



ur_fig_07. State UR publication-lag distribution

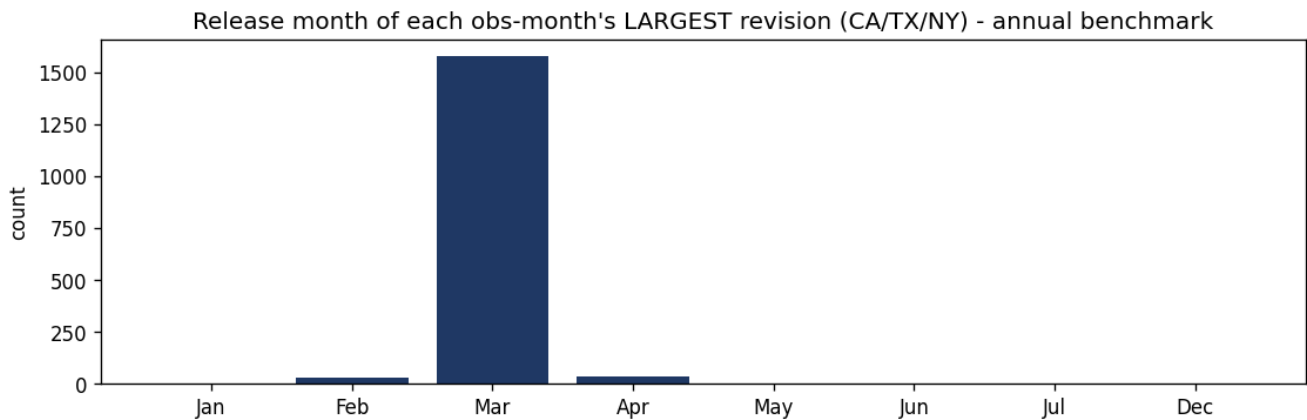


ur_fig_08. State UR release minus national jobs-report date

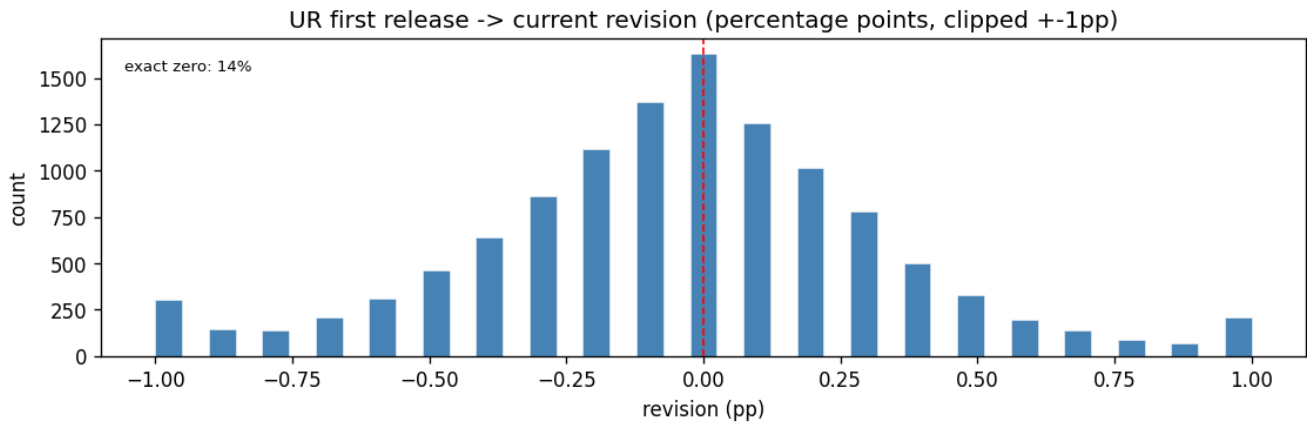


ur_fig_09. State UR first-release capture weekday

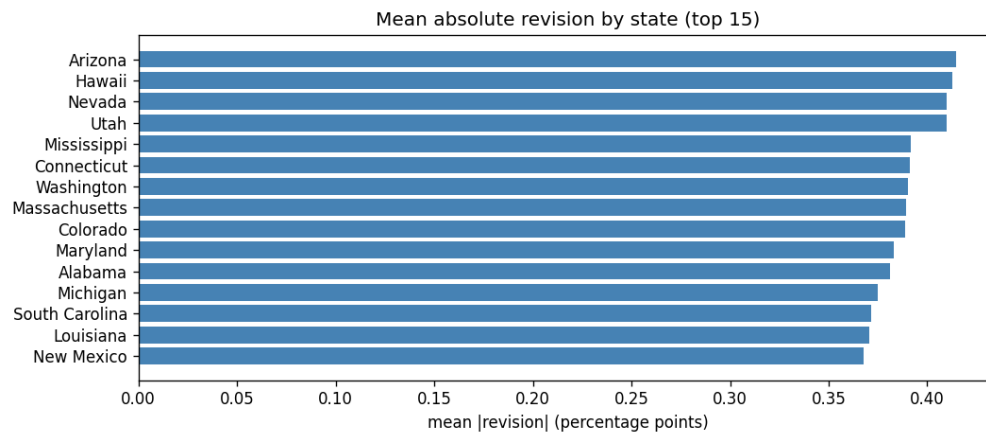
V4 - Revision behaviour



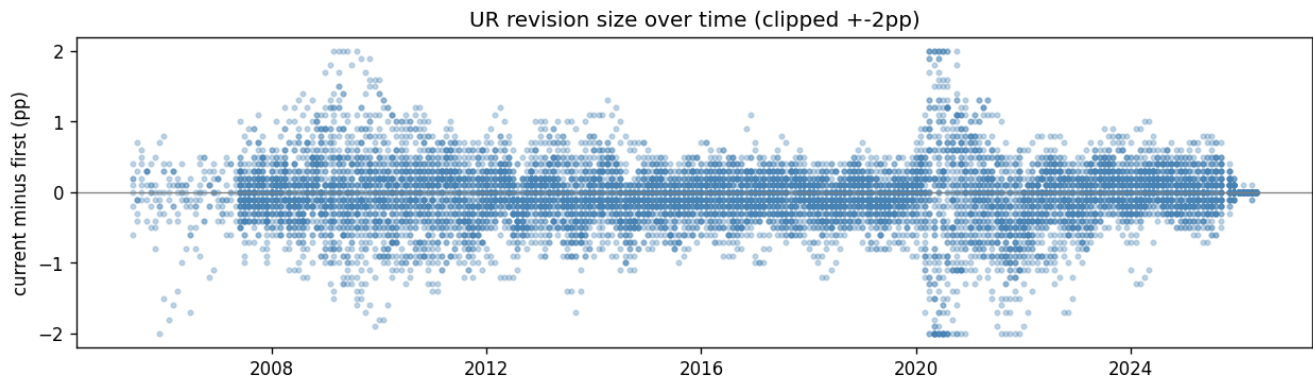
ur_fig_02. Release month of each obs-month's LARGEST revision (CA/TX/NY) - annual benchmark



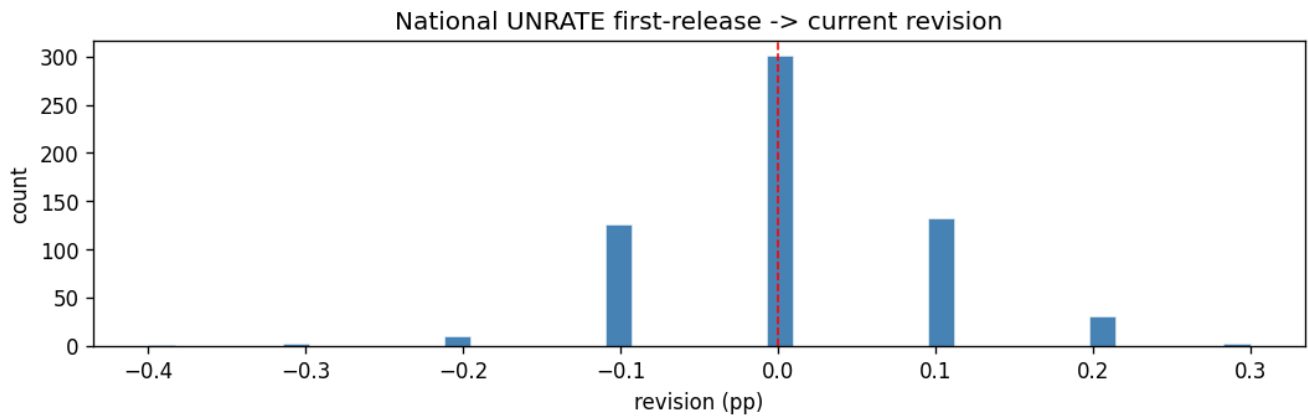
ur_fig_03. UR first release -> current revision (percentage points, clipped +-1pp)



ur_fig_10. Mean absolute revision by state (top 15)



ur_fig_11. UR revision size over time (clipped +-2pp)



ur_fig_12. National UNRATE first-release -> current revision

Nonfarm payrolls

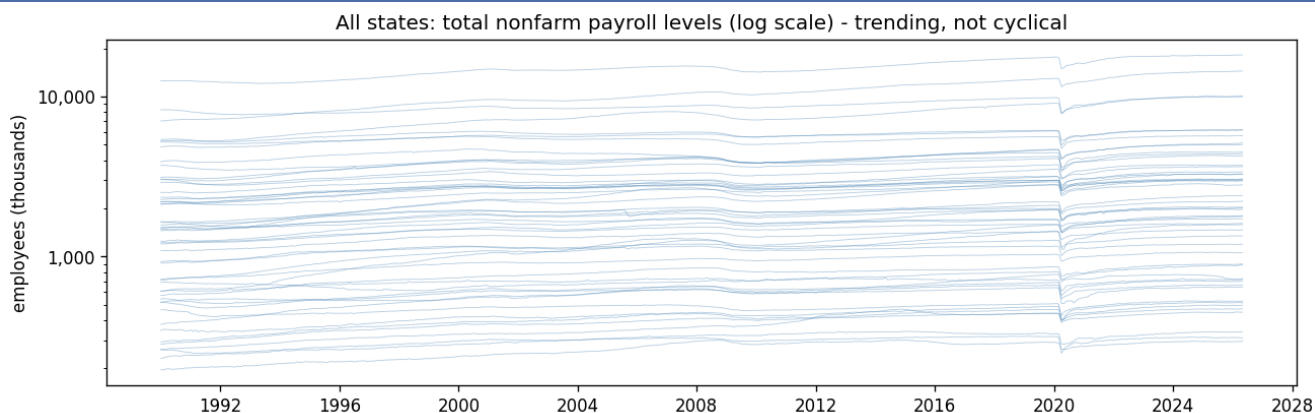
{ABBR}NA - monthly, SA level — 13 PASS / 4 WARN / 0 FAIL on 17 core checks

A heavily-revised, trending monthly level. Structure and calendar are clean; the defining feature is revision - ~99% of months change, driven by the annual QCEW benchmark (95% of the largest revisions land in the winter benchmark window).

ID	Check / question	Res	Metric
V1 - Structural integrity			
V1.1	Do we have all 51 states in every table?	PASS	{'vintage': 51, 'SA current': 51, 'pit': 51}
V1.2	Is (state, adjustment, month) a unique key?	PASS	vintage=0, current=0, pit=0
V1.3	Are key fields complete (no nulls)?	PASS	0 nulls
V1.4	Is every payroll level strictly positive and finite?	PASS	vintage bad=0, current bad=0
V1.5	Are the series lengths consistent across states?	PASS	min=437, max=437 months
V2 - Calendar & frequency			
V2.1	Are all observation dates month-starts?	PASS	0 non-first-of-month
V2.2	Are there missing months in any state?	PASS	0 gaps
V2.3	Do all states cover the same window?	PASS	start 1990-01-01..1990-01-01, end 2026-05-01..2026-05-01
V3 - Release calendar & point-in-time boundary			
V3.1	How far back can point-in-time be reconstructed?	WARN	first vintage 2005-07-22
V3.2	Does the vintage calendar match the jobs-report schedule?	WARN	Mon 3%, Tue 5%, Wed 5%, Thu 7%, Fri 80%
V3.3	Is the state publication lag ~50 days?	PASS	median 50d (IQR 48-53)
V3.4	Do state releases follow the national jobs report?	PASS	100.0% of state months released after national; median gap +14d
V3.5	Is the publication-lag regime stable over time?	WARN	median lag by year spans 48-66d (range 18d)
V4 - Revision behaviour			
V4.1	How often/how much does a first print get revised?	WARN	99.2% of months revised; median rev 0.55%, p90 1.63%, max 7.5%
V4.3	Is there a systematic bias in first prints?	PASS	mean revision +0.052% (first prints understate on average)
V4.2	Do revisions cluster at the annual benchmark?	PASS	95% of largest revisions land in Jan-Mar; modal release month = 3 (1249 of 1308 obs across CA/TX/NY)
V4.4	How large is the national benchmark (calibration)?	PASS	median +0.13%, IQR [-0.19%, +0.46%], max 1.4%

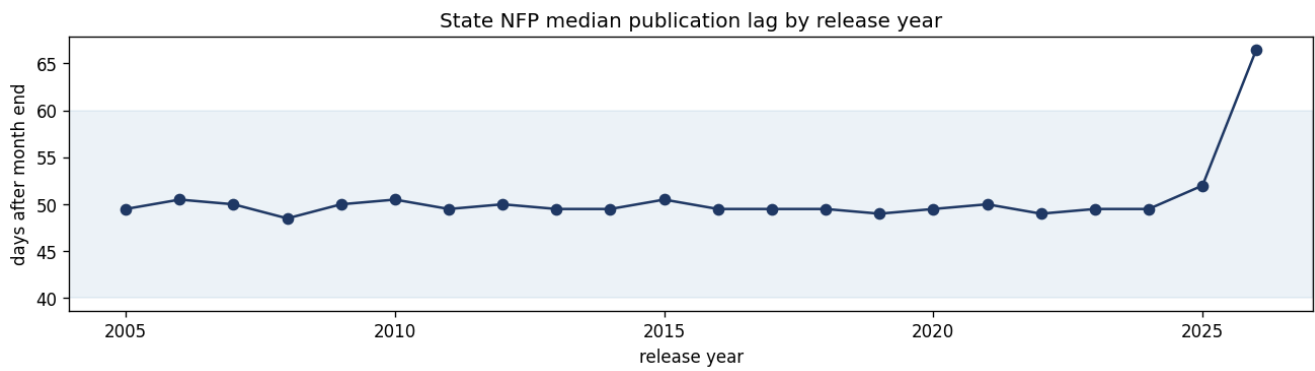
Core diagnostics

V1 - Structural integrity

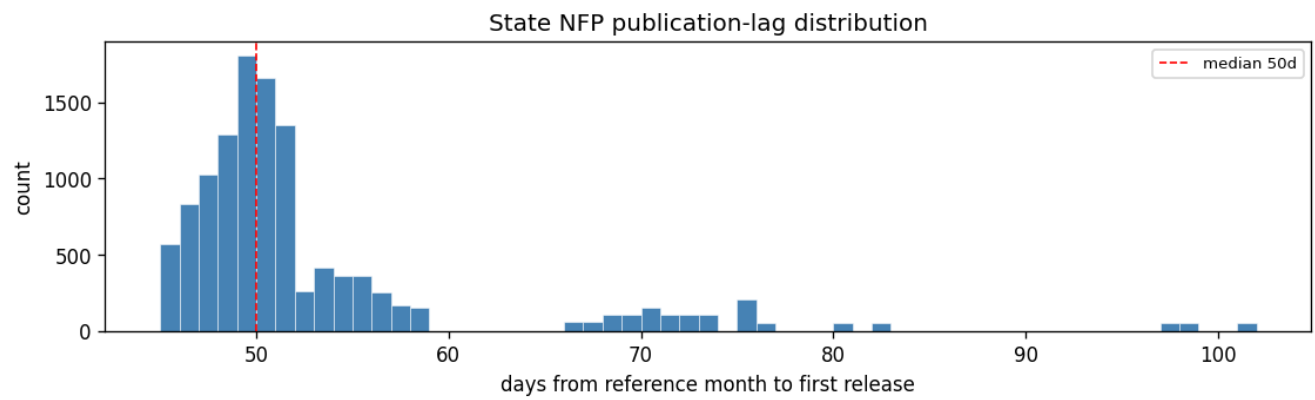


nfp_fig_15. All states: total nonfarm payroll levels (log scale) - trending, not cyclical

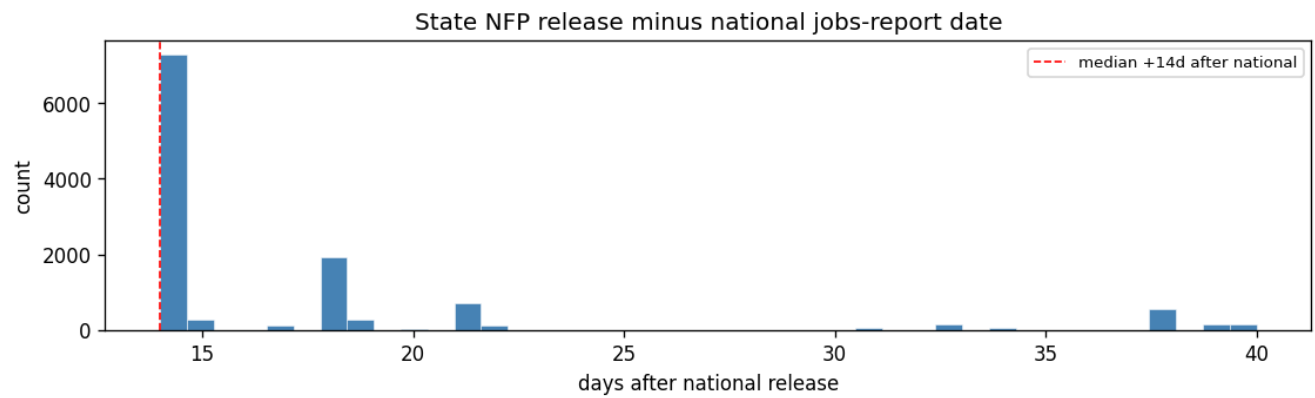
V3 - Release calendar & point-in-time boundary



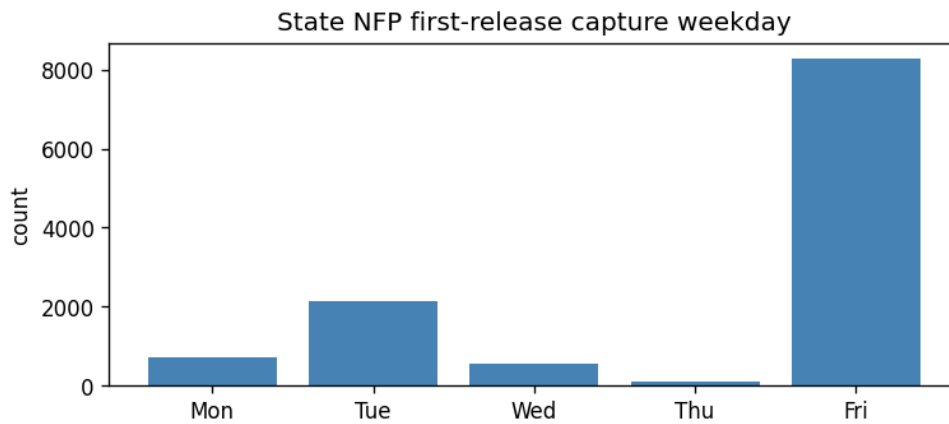
nfp_fig_01. State NFP median publication lag by release year



nfp_fig_09. State NFP publication-lag distribution

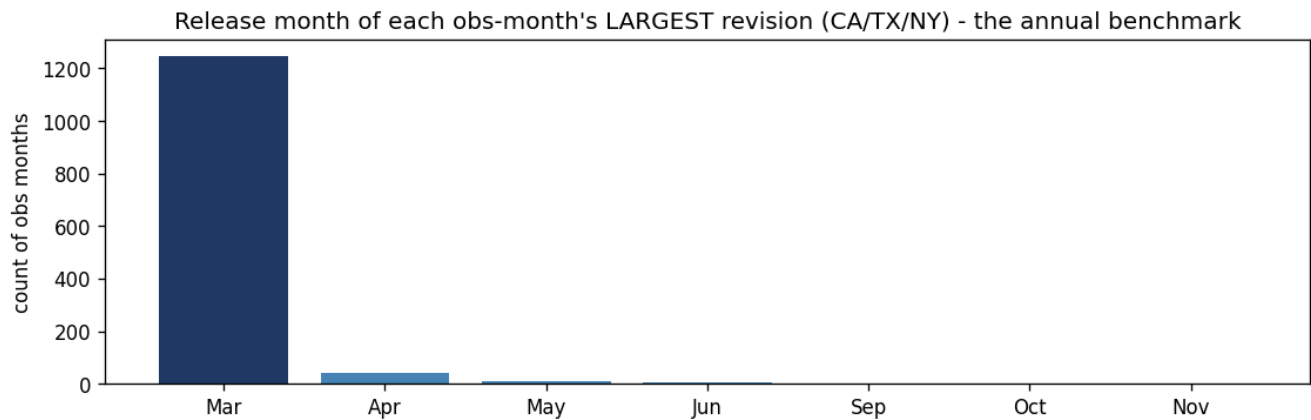


nfp_fig_10. State NFP release minus national jobs-report date

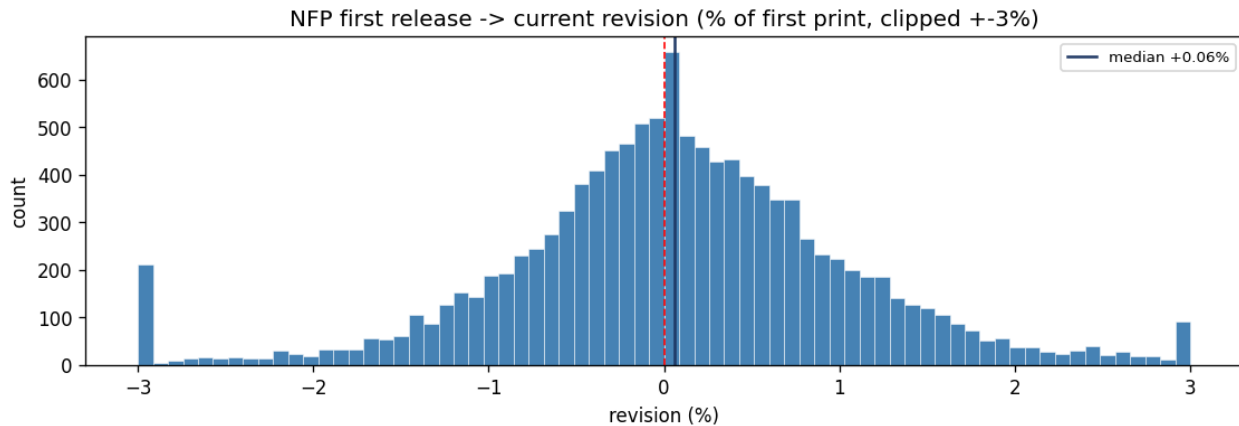


nfp_fig_11. State NFP first-release capture weekday

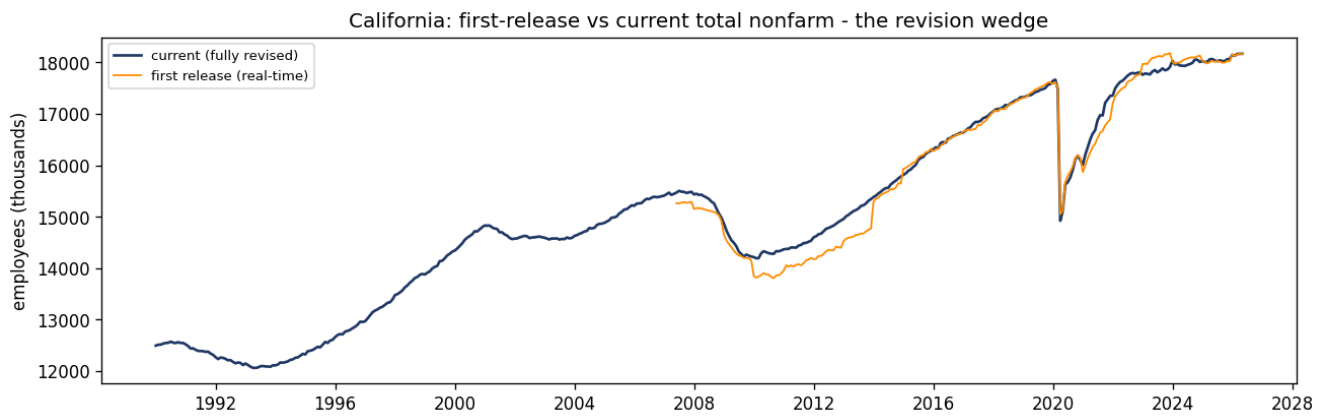
V4 - Revision behaviour



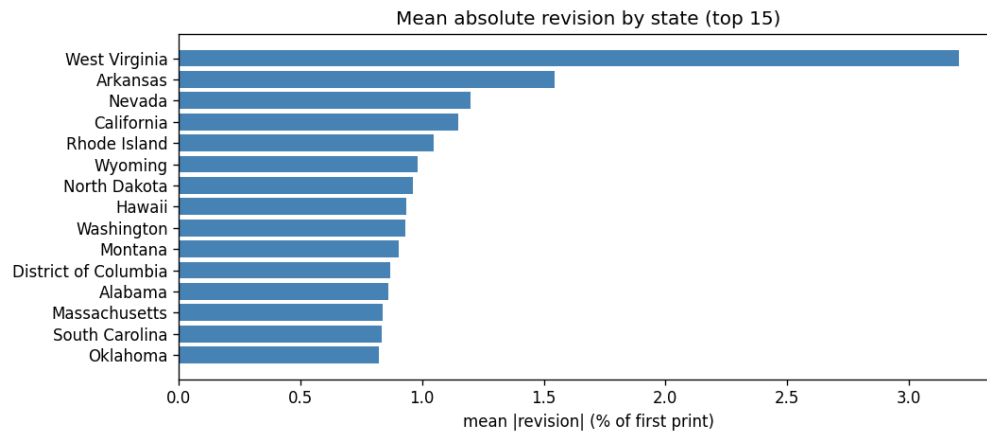
nfp_fig_02. Release month of each obs-month's LARGEST revision (CA/TX/NY) - the annual benchmark



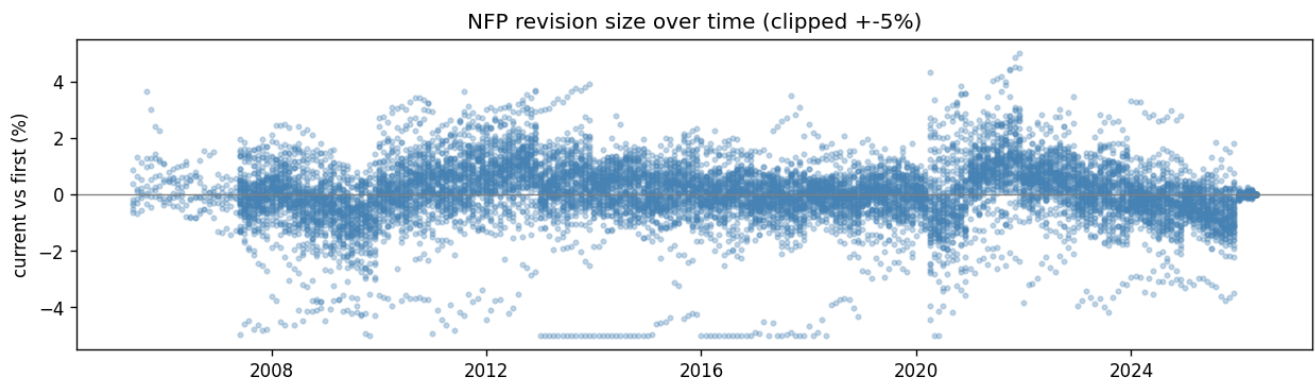
nfp_fig_03. NFP first release -> current revision (% of first print, clipped +/-3%)



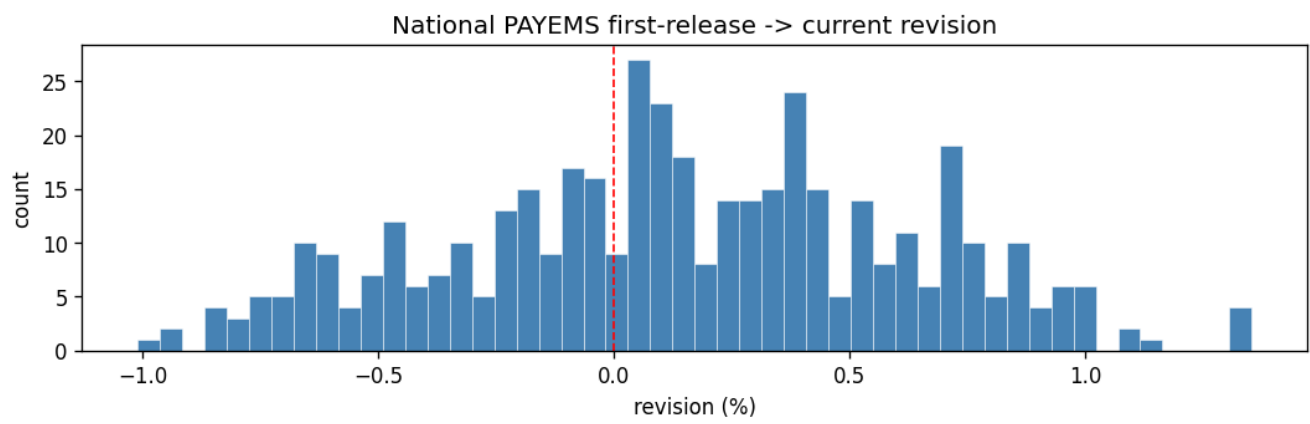
nfp_fig_04. California: first-release vs current total nonfarm - the revision wedge



nfp_fig_12. Mean absolute revision by state (top 15)



nfp_fig_13. NFP revision size over time (clipped +/-5%)



nfp_fig_14. National PAYEMS first-release -> current revision